

Figure 6.4.1.1 Quality of the brake fluid

In general the quality brake fluids such as DOT3, DOT4, DOT5, DOT5.1 are commonly used in modern vehicles. These brake fluids are certified by the Society of Automotive Engineers (S.A.E). The DOT3 type brake fluid is widely used in both cars and heavy vehicles.

Requirements of a brake fluid

The following are the important requirements of a brake fluid to be used in automobiles.

1. It should withstand high temperature. The boiling point of the brake fluid should be approximately 300°C.
2. Even at high temperatures the brake fluid should not lose it's viscosity and lubricity.
3. It should lubricate the various parts of the braking system.
4. It should not react and corrode with the rubber and metal parts of the braking system.
5. It should not lose its properties when it is stored for very long time (upto the maximum of 3 years).

Parts of the hydraulic braking system

The hydraulic braking system of an automobile consists of the following parts

1. Brake pedal
2. Master cylinder
3. Brake Fluid pipelines
4. Hose pipes
5. Wheel cylinder
6. Brake shoes
7. Brake linings
8. Return springs

The parts listed above are discussed in the following paragraphs.

Master cylinder and its type

Master cylinder acts as the heart of the hydraulic braking system. By pressing brake pedal, master cylinder supplies the required amount of the pressurized brake fluid to the wheel cylinders for expanding the brake shoes of the wheel cylinder. There are three different types of master cylinders used in automobiles. They are

1. Simple master cylinder
2. Tandem master cylinder
3. Center valve type master cylinder

Among the above types, the first two are mostly used in automobiles. The details of construction and working principle of the above two cylinders are discussed below.

Simple type master cylinder

Construction

The master cylinder is generally made up of cast iron material. The master cylinder consists of a reservoir tank for storing brake fluid and a compressor unit for operating the braking system. In these two parts are completely filled with brake fluid. In the top of the reservoir unit there

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